

Table 1. **World-model-evaluation deep-dive**: the 34 benchmarks of the world-model-evaluation lane, by sub-type – the P3 counterpart of the skill-building branch deep-dive. **Mode**: **Gen** evaluates generated video, **QA** video question-answering, **Pred** physical prediction. **Signal**: **Auto** computed/learned metric, **Acc** accuracy vs ground truth, **VOE** violation-of-expectation, **Human** human study. **Physical**: targets physical-law understanding, **✓** core, **~** adjacent, **✗** general. Generation-quality suites dominate and are physics-agnostic; only the physical-reasoning and physical-commonsense sub-types test physical law, and every lane benchmark scores open-loop, never in a closed loop.

Benchmark	Year	Mode	What it scores	Signal	Physical
Generation quality (multi-dimensional)					
EvalCrafter [22]	2023	Gen	17-metric text-to-video quality suite	Auto	✗
FETV [23]	2023	Gen	fine-grained text-to-video quality	Auto	✗
VBench [18]	2023	Gen	16 disentangled generation dimensions	Auto	✗
StoryBench [7]	2023	Gen	continuous story-visualisation quality	Auto	✗
DEVIL [21]	2024	Gen	content-dynamics quality of T2V	Auto	✗
T2V-CompBench [28]	2024	Gen	compositional text-to-video quality	Auto	✗
T2VScore [31]	2024	Gen	learned alignment + quality metric	Auto	✗
TC-Bench [12]	2024	Gen	temporal compositionality in generation	Auto	✗
VideoScore [16]	2024	Gen	learned automatic quality metric	Auto	✗
EWMBench [17]	2025	Gen	scene / motion / semantic quality	Auto	✗
VBench-2.0 [33]	2025	Gen	generation faithfulness	Auto	✗
Intuitive & physical reasoning					
IntPhys [27]	2018	Pred	intuitive physics from video	VOE	✓
Physion [5]	2021	Pred	physical outcome prediction from vision	Acc	✓
GRASP [19]	2023	QA	intuitive physics in video MLLMs	Acc	✓
Physion++ [29]	2023	Pred	prediction with online property inference	Acc	✓
ContPhy [34]	2024	QA	soft-body / fluid physical reasoning	Acc	✓
Physics-IQ [25]	2025	Gen	physical realism of generated video	Auto	✓
PhysBench [10]	2025	QA	physical-world understanding for VLMs	Acc	✓
PHYBench [26]	2025	QA	physical perception and reasoning	Acc	✓
IntPhys 2 [6]	2025	Pred	intuitive physics in complex scenes	VOE	✓
Physical commonsense in generated video					
PhyGenBench [24]	2024	Gen	physical-commonsense correctness in gen	Auto	✓
VideoPhy [3]	2024	Gen	physical commonsense in generated video	Human	✓
VideoPhy-2 [4]	2025	Gen	action-centric physical commonsense	Human	✓
PhyWorldBench [15]	2025	Gen	physical realism in text-to-video	Human	✓
IPV-Bench [1]	2025	Gen	impossible-video generation + understanding	Acc	✓

Table 1 (continued)

Benchmark	Year	Mode	What it scores	Signal	Physical
Counterfactual / causal / temporal					
CATER [14]	2019	QA	compositional-action temporal reasoning	Acc	✗
CLEVRER [32]	2019	QA	causal / counterfactual video reasoning	Acc	~
WorldPrediction [8]	2025	QA	counterfactual action recognition	Acc	✗
Video-language alignment / hallucination					
VideoCon [2]	2023	QA	robust video-language alignment	Auto	✗
VideoHalluciner [30]	2024	QA	hallucination in video-language models	Acc	✗
World-model prediction & world generation					
Vista [13]	2024	Gen	driving world-model generation	Auto	~
EVA-Bench [9]	2024	Pred	embodied video anticipation	Auto	~
WorldModelBench [20]	2025	Gen	video world-model prediction quality	Human	~
WorldScore [11]	2025	Gen	controllable world-generation quality	Auto	✗